

Information and score-covariance conventions for
the FIML robust GOF test:
structured vs unstructured, hessian vs h1,
and why the meat stays saturated

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Purpose

The robust reference law for the FIML likelihood-ratio statistic is the weighted mixture $T \Rightarrow \sum_{j=1}^{\text{df}} \lambda_j \chi_{1j}^2$ with $\lambda_j = \text{eig}_j\{U(V)\Omega\}$; see the companion note *Saturated FIML Information, MLR, and FMG under Non-normality* for the construction of V , Ω , the UT structure, and the Mplus two-trace bookkeeping. This note answers a narrower, recurring question: lavaan exposes *three* computational switches that all sound like “structured vs saturated,” and they are routinely conflated. We

1. separate the three axes and state, in a clean (J, K, Δ) notation, what each one varies (Section);
2. record that the **hessian**/**h1** axis has *opposite* correct answers for standard errors and for the test (Section);
3. give the whitening identity that localizes all non-normality in the meat, and correct an overclaim about *why* the meat is held saturated: the case is finite-sample and numerical, not asymptotic size (Section);
4. document a positive-definiteness hazard in the structured- V path and audit what magmaan does and does not guard (Section).

The headline conclusions: magmaan uses observed information, builds U in saturated (η) coordinates so it is implicitly on the **h1** side, and holds the meat Ω at the saturated estimate. Only the U -side curvature is offered

structured or saturated, and the structured option is the one with the failure modes.

Setup and notation

Let $\xi = (\mu, \text{vech } \Sigma)$ be the saturated (H_1) parameters, $\ell(\xi; x) = \log \phi_R(x; \xi)$ the per-observation observed-data normal log-likelihood for missingness pattern R , and $u(\xi; x) = \partial \ell / \partial \xi$ the saturated score. Write

$$J = -\mathbb{E}\{D_\xi^2 \ell(\xi^*)\}, \quad K = \text{Cov}\{u(\xi^*)\}, \quad \Omega = J^{-1} K J^{-1},$$

all at the least-false ξ^* ; Ω is the sandwich asymptotic covariance of the saturated estimate $\hat{\xi}$. Let $\Xi : \theta \mapsto \xi$ be the restricted SEM map, $\Delta = \partial \Xi / \partial \theta'$ (projected into the equality-constraint tangent space when present), and

$$U(V) = V - V \Delta (\Delta' V \Delta)^{-1} \Delta' V.$$

Every one of J , K , V must be evaluated at some point; the candidates are the saturated MLE $\hat{\xi}$ and the restricted point $\xi(\hat{\theta})$. Finite-sample magmaan units ($V_N \sim N$, $\hat{\Gamma}_{\text{mis}} \sim 1/N$) are as in the companion note and do not affect any eigenvalue here.

Three computational axes

The four-cell tables of Savalei and Rosseel [1] cross three *independent* binary choices. In our notation:

A. information = expected / observed. How J (their A_β) is formed.

Observed is the genuine averaged second derivative $-\frac{1}{n} \sum_i D_\xi^2 \ell_{x_i}$, which uses the *sample* per-pattern moments $\bar{y}_j, S_j$ (their Eq. 5 form, carrying $h_j = \Sigma_j^{-1}(\bar{y}_j - \mu_j)$). *Expected* replaces the sample moments by the model's own, giving the clean block-diagonal Fisher form $W(\hat{\xi})$ (their Eq. 8). magmaan uses *observed*: the analytic routine `fiml_saturated_hessian_analytic_block` builds exactly the sample-moment Hessian (the $\bar{y}_j - \mu_j$ and S_j terms are explicit), not the Fisher form.

B. h1.information = structured / unstructured. The *evaluation point* $\hat{\xi}$. *Unstructured* evaluates at the saturated MLE $\hat{\xi}$; *structured* at the model-implied $\xi(\hat{\theta})$. This is the only axis magmaan exposes, as `estimate::FIMLH1Information::\{Saturated, Structured\}`, and it acts on the U -side curvature V only.

C. observed.information = hessian / h1. A θ -space choice: whether the structural information includes the chain-rule curvature term. With d_β the saturated gradient and $H = \partial(\vec{\Delta}')/\partial\theta'$,

$$\underbrace{A_\theta = \Delta' J \Delta - (d'_\beta \otimes I)H}_{\text{hessian}}, \quad \underbrace{A_{\theta, \text{h1}} = \Delta' J \Delta}_{\text{h1}}.$$

magmaan never forms the numeric θ -Hessian; it builds U in ξ -space via $\Delta' V \Delta$, so it is implicitly **h1**.

Remark (The names collide two axes). “h1” in axis C names the *model whose information matrix is sandwiched* ($\Delta' A_\beta \Delta$), not an evaluation point; the evaluation point is axis B. So **observed.information="h1"** and **h1.information="structured"** are orthogonal, despite both saying “h1”. “hessian” is equally unfortunate: both objects in axis C are Hessians. The collision is the source of the recurring confusion.

Hessian vs h1: opposite verdicts for SEs and the test

Lemma 1. *The correction $(d'_\beta \otimes I)H$ is zero at $\tilde{\xi}$ (saturated first-order condition $d_\beta(\tilde{\xi}) = 0$) and is $O_p(\text{misfit})$ at $\xi(\hat{\theta})$, vanishing under H_0 . Hence **hessian** and **h1** coincide for unstructured evaluation, and differ for structured evaluation only off H_0 .*

Standard errors: hessian is the consistent bread. Under misspecification the Huber–White bread must be the genuine Hessian $A_\theta = -D_\theta^2 \ell(\hat{\theta})$ [6]. That is the **hessian** object; the **h1** object drops a term nonzero off H_0 . So $\hat{A}_\theta$ is consistent for A_θ^* whatever the truth, while $\Delta' \hat{A}_\beta \Delta$ is consistent only under H_0 [1]. For SEs, **h1** is the approximation.

The test: h1 is the exact projector. For the GOF test the θ -space information enters only as the middle inverse of U . Compare

$$U_{\text{h1}} = V - V \Delta (\Delta' V \Delta)^{-1} \Delta' V, \quad U_{\text{hess}} = V - V \Delta A_\theta^{-1} \Delta' V.$$

U_{h1} is an exact residual operator of rank $\text{df} = \dim \xi - \text{rank } \Delta$ by construction; U_{hess} has that rank only asymptotically and only under H_0 [1, Table 6]. Off H_0 the hessian middle leaks spurious nonzero eigenvalues into the mixture. magmaan’s ξ -space construction is U_{h1} ; the post-hoc check that the largest projected-out eigenvalue is ≈ 0 (`fiml_ugamma_spectrum_impl`) is meaningful precisely because the operator is the exact projector.

Remark. So `observed.information="hessian"` is not “wrong”; it is the right bread for misspecification-robust SEs and the wrong shape for the GOF test (rank impurity for no gain). We do the test, we are implicitly `h1`, and that is the correct call.

Why the meat Ω stays saturated

Lemma 2 (Whitening). *If the U -side curvature equals the meat bread, $V = J$, then with $J = J^{1/2}J^{1/2}$,*

$$\text{eig}\{U(J)\Omega\} = \text{eig}\{(I-P)J^{-1/2}KJ^{-1/2}\}, \quad P = J^{1/2}\Delta(\Delta'J\Delta)^{-1}\Delta'J^{1/2},$$

where P is the orthogonal projector onto $\text{range}(J^{1/2}\Delta)$ of rank $\text{rank } \Delta$.

Proof. $U(J) = J^{1/2}(I-P)J^{1/2}$, so $U(J)\Omega = J^{1/2}(I-P)J^{-1/2}KJ^{-1/2}$; conjugating by $J^{-1/2}$ preserves eigenvalues and yields $(I-P)J^{-1/2}KJ^{-1/2}$. $\square$

Lemma 2 says all non-normality lives in the *whitened meat* $J^{-1/2}KJ^{-1/2}$ projected onto the rank-df residual space: under the information equality $K = J$ it is the identity, every residual eigenvalue is one, and $T \Rightarrow \chi_{\text{df}}^2$; the spectrum spreads exactly as K departs from J . The meat is the matrix one must not get wrong.

Remark (Size is fine either way; the case is finite-sample). Under H_0 both the saturated meat $\tilde{K}$ and a (centered) structured meat $\hat{K}$ are consistent for the null K , so both give asymptotically correct size; under a fixed alternative the test is consistent regardless of the meat. There is therefore *no* asymptotic argument either way, and the earlier claim that the saturated meat is “the only choice consistent under the alternative” is too strong. The real case for the saturated meat is finite-sample and numerical:

1. $\tilde{\xi} \rightarrow \xi^*$ whether or not H_0 holds (the saturated model is never misspecified), and at $\tilde{\xi}$ the saturated scores are mean-zero by the first-order condition, so $\tilde{K} = \frac{1}{4} \sum_i u(\tilde{\xi})u(\tilde{\xi})'$ is a clean variance;
2. the uncentered structured meat at $\xi(\hat{\theta})$ adds $\bar{s}\bar{s}'$, the squared misfit gradient, which is 0 only in the limit under H_0 and inflates the meat in finite samples and under misfit;
3. the Satorra–Bentler convention holds Ω saturated and varies only $U(V)$ [2, 3], and the complete-data evidence mildly prefers saturated information [7, 1].

Remark (magmaan’s structured cell is a hybrid). Lemma 2 needs $V = J$. magmaan’s **Structured** option uses $V = V(\xi(\hat{\theta}))$ for U while keeping the meat bread at $\tilde{\xi}$; the clean whitening then fails, which is exactly the Satorra–Bentler “structured U^0 , unstructured Ω^0 ” cell rather than a fully structured construction.

Positive-definiteness hazard of structured V

Lemma 3. *$V(\tilde{\xi})$ is positive definite (observed information at the saturated optimum, by second-order optimality). $V(\xi(\hat{\theta}))$ is the saturated observed Hessian at a non-optimal point; off H_0 its data terms in $\bar{y}_j - \mu_j(\hat{\theta})$ need not be dominated and V can be indefinite.*

When V is indefinite, $U(V)$ loses its residual-projector meaning, $\Delta'V\Delta$ may be indefinite, and a retained eigenvalue can turn negative even though the mixture $\sum_j \lambda_j \chi_{1j}^2$ assumes $\lambda_j \geq 0$. The default **Saturated** path is unaffected by Lemma 3; the hazard is confined to the opt-in **Structured** path, and it bites precisely where the model is misspecified (the regime that a GOF simulation targets).

What magmaan does and does not guard. As of this writing the structured path crosses its fingers:

- No definiteness test on V . `fiml_structured_h1_information` only Cholesky-tests the model-implied Σ_{oo} (always PD), which says nothing about the assembled V .
- `invert_symmetric` tries LLT and on failure *silently* falls back to LDLT, which succeeds for any non-singular symmetric matrix; an indefinite $\Delta'V\Delta$ is inverted with no error or signal, only an outright singular one fails.
- The rank sanity check inspects a *single* boundary eigenvalue and does not check the sign of the retained df eigenvalues that feed the FMG/pEBA mixture.

Status. Saturated is the default and the FMG-spectrum convention. Structured-h1 is not asymptotically advantageous, is PD-unsafe, and is the combination that reproduces lavaan-MLR’s bread (so removing it would end exact MLR bread reproduction from magmaan, though an MLR baseline run via lavaan

is unaffected). An evaluation is queued in the backlog: under the experiment-23 misspecified conditions, measure how often structured $V/\Delta'V\Delta$ goes indefinite and whether structured-h1 ever beats saturated on Type-I or power; if it never helps, drop the option and keep **Saturated** as the sole FIML-FMG path.

References

- [1] Savalei, V., & Rosseel, Y. (2022). Computational options for standard errors and test statistics with incomplete normal and nonnormal data in SEM. *Structural Equation Modeling*, 29(2), 163–181. doi:10.1080/10705511.2021.1877548.
- [2] Satorra, A., & Bentler, P. M. (1994). Corrections to test statistics and standard errors in covariance structure analysis. In *Latent Variables Analysis* (pp. 399–419). Sage.
- [3] Satorra, A. (2000). Scaled and adjusted restricted tests in multi-sample analysis of moment structures. In *Innovations in Multivariate Statistical Analysis* (pp. 233–247). Kluwer.
- [4] Yuan, K.-H., & Bentler, P. M. (2000). Three likelihood-based methods for mean and covariance structure analysis with nonnormal missing data. *Sociological Methodology*, 30, 165–200. doi:10.1111/0081-1750.00078.
- [5] Browne, M. W. (1984). Asymptotically distribution-free methods for the analysis of covariance structures. *British Journal of Mathematical and Statistical Psychology*, 37, 62–83. doi:10.1111/j.2044-8317.1984.tb00789.x.
- [6] White, H. (1982). Maximum likelihood estimation of misspecified models. *Econometrica*, 50(1), 1–25. doi:10.2307/1912526.
- [7] Xia, Y., Yung, Y.-F., & Zhang, W. (2016). Evaluating the selection of normal-theory weight matrices in the Satorra–Bentler correction of chi-square and standard errors. *Structural Equation Modeling*, 23, 585–594. doi:10.1080/10705511.2016.1141354.
- [8] Imhof, J. P. (1961). Computing the distribution of quadratic forms in normal variables. *Biometrika*, 48, 419–426. doi:10.2307/2332763.